**Quiz – 1 (Contributes towards Class Participation)**

**Total marks – 20**

1. For the experiment of drawing 5 cards out of a deck of 52 cards, what is the sample space? If X is defined to be the number of hearts in the 5 cards drawn, what is the distribution of X? What is the mean of X? [2 Marks]

2. Given a two sided coin. Using this coin, how will you simulate an n-sided coin:
   a. When n is a power of 2?
   b. When n is not a power of 2? [3 Marks]

3. A coffee can contains some black beans and white beans. The following process is repeated as long as possible. Randomly select two beans from the can. If they have the same color, throw them out, but put another black bean in. (Enough extra black beans are avail- able to do this.) If they are different colors, put the white one back into the can and throw the black one away. Prove that the process stops with exactly one bean in the can. Can you claim anything about the color of the final bean ? [3 Marks]

4. An array A has n/4 copies of a particular unknown 'x'. Every other element in A has at most n/8 copies. Present an O(log n) time Monte Carlo algorithm to identify x. [3 Marks]

5. Give both a recursive and an iterative algorithm to compute the binomial coefficient $\binom{n}{m}$. [3 Marks]

6. Find the complexity of the following using any method discussed in class: [6 Marks]
   a. $T(n) = 2\,T(n/4) + \sqrt{n}$
   b. $T(n) = T(n-1) + 1/n$
   c. $T(n) = T(n/3) + T(n/6) + n$